NED University of Engineering and Technology
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GPU Sharing Mechanisms for Multi-Tenant Inference Serving

Utilization, Latency Isolation and Engineering Trade-offs

Operating Systems CEP, Topic23
Systematic Review Evidence and Analysis

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Current evidence checkpoint, 16 October 2026. All canonical records have a first automated triage recommendation. Full abstract verification and final eligibility remain incomplete. Fifteen assessed report forms are not fifteen final included studies. This report preserves those distinctions and the authorized AI-only review deviation.

Supporting files: compact review workbook; separate QA and data-extraction tables; record-keyed CSV logs.

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1 Background

GPU sharing is an operating-system and runtime decision about how concurrent clients use compute, memory and scheduling capacity on one physical accelerator. Throughput and model density can improve while request tails, interference, memory pressure or isolation worsen. The relevant comparison is therefore conditional on workload, hardware and baseline, rather than a single system-wide ranking.

The assigned families are temporal sharing, MPS-style concurrency, MIG partitions, API interception/remoting and hybrids. Product documentation supplies definitions, not measured inference gains. The reviewed primary reports supply the experimental observations.

Context and review methods: [1, 2, 3, 4, 5, 6].

2 Engineering problem and scope

Choose a sharing mechanism for concurrent inference on one physical GPU under specified latency, utilization, density and isolation requirements. Multi-GPU papers are eligible only when their local co-tenant inference evidence is separable. Mixed training/inference reports are assessed using inference results rather than relabelling training throughput. Remoting is relevant where the endpoint GPU is shared and tenant effects are evaluable.

Decision constraint	Implication
Tail-latency objective	A mean or isolated kernel latency cannot establish request P99/P98 compliance.
Memory and model density	Additional functions may require model swapping or runtime sharing; transfer and trust costs must be retained.
Isolation boundary	Performance fairness, memory separation, fault containment and security are different claims.
Deployment environment	GPU generation, compiler, driver hooks, runtime changes and host CPU demand affect applicability.

3 Objectives and research questions

The objective is a traceable conditional recommendation that relates a mechanism to a workload and a sacrifice. The submitted protocol remains unchanged; later deviations are documented separately.

RQ	Review question
RQ1	What time-slicing, MPS, MIG and API remoting mechanisms are evaluated for concurrent inference on one GPU, and how do sharing granularity, layer and isolation boundary classify them?
RQ2	How do studies measure utilization, throughput, density, average/tail latency, predictability and interference, and how comparable are their experimental methods?
RQ3	Under what GPU, workload, batch, tenant mix and configuration conditions are changes in utilization, throughput, density and latency reported?

RQ	Review question
RQ4	What resource and fault isolation properties, trade-offs, implementation costs and conflicting results are reported, and what limits the evidence?
RQ5	Which approach or combination is supported for specified inference-serving conditions and priorities, with what confidence and alternatives?

4 Review methodology

Occurrence, canonical record, report, study and outcome are separate units. Raw exports retain source file, position and hash. Duplicate links alter the canonical screening queue without removing occurrences or earlier votes. Full-text criteria, QA rationales and outcome rows carry report IDs; final unique studies require report-to-study reconciliation.

On 16 October 2026, the user authorized AI-only continuation without waiting for manual returns. Automated triage and source-based AI assessment therefore proceed as an explicit departure from the planned two-reviewer process. The process does not claim independent human agreement. Original votes/scores remain available; unperformed second/reconciled fields stay blank.

The new triage rules exclude only explicit review/non-primary or specified non-GPU-domain signals. Other unresolved metadata is retained as uncertain. This closes first-pass coverage, not full-abstract verification or final eligibility. Accessible conference/author PDFs are assessed in bounded batches and saved with retrieval hashes. A DOI alone does not establish source access or eligibility.

5 Current accounting and review status

Stage	Count / interpretation
Imported occurrences	4189
Applied duplicate links	114 =111 earlier links plus3 publisher-verified identities
Canonical records	4075
First automated recommendations	4075
Preliminary exclusion proposals	1441
Retain or uncertain	2634
Source-assessed reports	17
Assessed full-text exclusions	2; AI EC2 judgments, not complete census
QA/extraction report forms	15; not final studies
Q1-Q10 item rows	150
Outcome rows	45; not study counts
Final included reports/studies	Pending; no numerical final claim

The three applied identity links are HealthServe DOI identity and two DOI-less occurrences whose publisher URLs match DARIS and CRONUS. The removed canonical rows remain in screening_merged_history.csv; all vote events remain in title_votes.csv. Other version or companion proposals remain separate. Source counts are parsed occurrences, not displayed search hits.

6 Protocol and documented deviations

Deviation	Handling / consequence
Attributed IEEE/Springer searches	Shared Group A runs are reused by authorization and attributed; they are not newly executed Group B searches.
ScienceDirect fallback queries	The master expression was rejected; available fallback exports are active. Complete execution of every planned lane is not demonstrated.
Search fields and filters	Executed fields and URL facets govern provenance. Full-text indexing and an open-access temporal filter can change sensitivity and specificity.
Historical count discrepancies	ACM1600displayed/1603claimed exports versus1582parsed numbered exports; ScienceDirect MPS517displayed/563exported remain unresolved.
AI-only continuation	Authorized departure from independent two-reviewer plan; no human votes invented.
Version uncertainty	ParvaGPU accessed through arXivv1; final publication equivalence unresolved. REEF companion relationship remains open.
Incomplete final selection	First-pass triage and assessed source slice do not provide final PRISMA stages or final study count.

7 Academic databases and provenance

Source	Occurrences	Linked repeats	Canonical attribution
ACM	1582	30	1552
IEEE	102	9	93
Springer	587	3	584
ScienceDirect	1898	67	1831
Supplementary	20	5	15

ACM, IEEE Xplore, Springer and ScienceDirect are represented in the active imports. Supplementary sources are tracked separately. Cross-database duplicates are attributed to the surviving canonical record; per-source remaining counts are not independent studies. Four represented platforms do not establish a reproducible complete four-database search.

8 Search strategy and executed query tables

The planned expressions are preserved in planned_queries.csv. The following table reproduces the executed or operator-reported query register, including its uncertainty. Displayed hits and exported/parsed occurrences remain distinct. Full filenames and provenance are retained in the workbook search/export logs; unknown historical fields remain unknown.

IEEE-MPS

Field	Executed/register evidence
Platform / status	IEEE Xplore; attributed shared executed search; not rerun by Group B
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND (MPS OR "multi-process service")
Fields / filters	All metadata (Search by All); not field-qualified Command Search; Publication years2020-2026; date/language/type checked during screening
Date / timezone	2026-10-11; time not recorded; timezone unknown
Displayed / claimed exported	25 / 25
Parsed / active	25 / 25
Evidence / gaps	evidence/imports/group-a/imported_records.json; Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer

IEEE-MIG

Field	Executed/register evidence
Platform / status	IEEE Xplore; attributed shared executed search; not rerun by Group B
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND (MIG OR "multi-instance GPU" OR "GPU partitioning")
Fields / filters	All metadata (Search by All); not field-qualified Command Search; Publication years2020-2026; date/language/type checked during screening
Date / timezone	2026-10-11; time not recorded; timezone unknown
Displayed / claimed exported	23 / 23
Parsed / active	23 / 23
Evidence / gaps	evidence/imports/group-a/imported_records.json; Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer

IEEE-TEMPORAL

Field	Executed/register evidence
Platform / status	IEEE Xplore; attributed shared executed search; not rerun by Group B
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching")

Field	Executed/register evidence
Fields / filters	All metadata (Search by All); not field-qualified Command Search; Publication years2020-2026; date/language/type checked during screening
Date / timezone	2026-10-11; time not recorded; timezone unknown
Displayed / claimed exported	45 / 45
Parsed / active	45 / 45
Evidence / gaps	evidence/imports/group-a/imported_records.json; Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer

IEEE-API

Field	Executed/register evidence
Platform / status	IEEE Xplore; attributed shared executed search; not rerun by Group B
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields / filters	All metadata (Search by All); not field-qualified Command Search; Publication years2020-2026; date/language/type checked during screening
Date / timezone	2026-10-11; time not recorded; timezone unknown
Displayed / claimed exported	9 / 9
Parsed / active	9 / 9
Evidence / gaps	evidence/imports/group-a/imported_records.json; Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer

SPR-MASTER

Field	Executed/register evidence
Platform / status	Springer Nature Link; attributed shared executed search; not rerun by Group B
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields / filters	General search/full-text indexed collection; English; Research article OR Conference paper;2020-2026; manual cutoff

Field	Executed/register evidence
Date / timezone	2026-10-11; time not recorded; timezone unknown
Displayed / claimed exported	587 / 587
Parsed / active	587 / 587
Evidence / gaps	evidence/imports/group-a/imported_records.json; Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer

ACM-PILOT

Field	Executed/register evidence
Platform / status	ACM Digital Library; attributed shared executed search; not rerun by Group B
Exact query	GPU AND inference AND MPS
Fields / filters	not_validated; not_applied
Date / timezone	2026-10-11; time not recorded; timezone unknown
Displayed / claimed exported	UNKNOWN / 0
Parsed / active	0 / 0
Evidence / gaps	evidence/imports/group-a/imported_records.json; Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer

SD-PILOT

Field	Executed/register evidence
Platform / status	ScienceDirect; attributed shared executed search; not rerun by Group B
Exact query	GPU AND inference AND MPS
Fields / filters	not_validated; not_applied
Date / timezone	2026-10-11; time not recorded; timezone unknown
Displayed / claimed exported	UNKNOWN / 0
Parsed / active	0 / 0
Evidence / gaps	evidence/imports/group-a/imported_records.json; Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer

SUP-SELECTED

Field	Executed/register evidence
Platform / status	Supplementary public web/venue/author lookup; attributed shared executed search; not rerun by Group B
Exact query	Named-paper lookups and official venue/author retrieval; see supplementary_discovery.csv
Fields / filters	Public metadata/full texts; Same protocol eligibility
Date / timezone	2026-10-11; time not recorded; timezone unknown
Displayed / claimed exported	UNKNOWN / 20
Parsed / active	0 / 0
Evidence / gaps	evidence/imports/group-a/imported_records.json; Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer

ACM-MASTER

Field	Executed/register evidence
Platform / status	ACM Digital Library; reported master query; master and numbered exports not fully mapped
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields / filters	All Fields; 2020 - 2026
Date / timezone	2026-10-14 (operator reported); 14 October 2026, 10:30 PM Asia/Karachi
Displayed / claimed exported	1600 / 1603
Parsed / active	1829 / 0
Evidence / gaps	evidence/incoming_shared/acm_master/ACM_MASTER_01.bib;evidence/incoming_shared/acm_master/ACM_MASTER_02.bib; 1600 displayed and 1603 exported claimed; 1829 overlapping master entries with 1387 distinct DOI/title keys; 68 staged unmatched candidates; original 79-file mapping unresolved

SD-TEMPORAL

Field	Executed/register evidence
Platform / status	ScienceDirect; historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND ("time slicing" OR "time sharing" OR "context switching")
Fields / filters	Find articles with these terms; 2020 - 2026

Field	Executed/register evidence
Date / timezone	2026-10-14 (operator reported); 14 October 2026, 11:45 PM Asia/Karachi
Displayed / claimed exported	1053 / 1053
Parsed / active	1053 / 1053
Evidence / gaps	evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_01.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_010.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_011.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_02.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_03.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_04.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_05.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_06.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_07.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_08.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_09.ris; TEMPORAL URL has openaccess facet omitted from filter text; 1053 parsed agrees with reported count; exact pagination and operator unknown

SD-MPS

Field	Executed/register evidence
Platform / status	ScienceDirect; historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND (MPS OR "multi-process service")
Fields / filters	Find articles with these terms; 2020 - 2026
Date / timezone	2026-10-14 (operator reported); 14 October 2026, 11:52 PM Asia/Karachi
Displayed / claimed exported	517 / 563
Parsed / active	563 / 563
Evidence / gaps	evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_01.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_02.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_03.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_04.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_05.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SSD_MPS_06.ris; 563 parsed versus 517 displayed; export claim 563; operator and exact pagination unknown

SD-MIG

Field	Executed/register evidence
Platform / status	ScienceDirect; historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND (MIG OR "multi-instance GPU" OR "GPU partitioning")
Fields / filters	Find articles with these terms; 2020 - 2026
Date / timezone	2026-10-14 (operator reported); 14 October 2026, 11:55 PM Asia/Karachi
Displayed / claimed exported	241 / 241
Parsed / active	241 / 241
Evidence / gaps	evidence/incoming_shared/sciencedirect_2026-10-16/SD_MIG_01.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MIG_02.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MIG_03.ris; Parsed count agrees with reported count; identity and page completeness unverified; operator unknown

SD-API

Field	Executed/register evidence
Platform / status	ScienceDirect; historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND ("API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields / filters	Find articles with these terms; 2020 - 2026
Date / timezone	2026-10-14 (operator reported); 14 October 2026, 11:56 PM Asia/Karachi
Displayed / claimed exported	41 / 41
Parsed / active	41 / 41
Evidence / gaps	evidence/incoming_shared/sciencedirect_2026-10-16/SD_API_01.ris; Parsed count agrees with reported count; identity and page completeness unverified; operator unknown

ACM-NUMBERED-UNMAPPED

Field	Executed/register evidence
Platform / status	ACM Digital Library; actual supplied exports; run mapping unknown
Exact query	unknown
Fields / filters	unknown; unknown
Date / timezone	unknown; unknown
Displayed / claimed exported	unknown / unknown
Parsed / active	1582 / 1582

Field	Executed/register evidence
Evidence / gaps	evidence/imports/acm/*.bib; exact files in import_manifest.csv; Number 3 absent in names 1-80; this is not proof of missing search page; do not assign to master query without mapping evidence

9 Literature period

The main period is2020 through the actual 2026 search date for each run. Shared logs indicate 11 October 2026; ACM reports 14 October 2026 coverage. Those dates are not treated as one universal cutoff. Later primary-paper retrieval is source assessment, not a claim that an earlier database run covered later publications. Older seminal exceptions require an individually documented justification; they are not silently mixed into the main window.

10 Inclusion and exclusion criteria

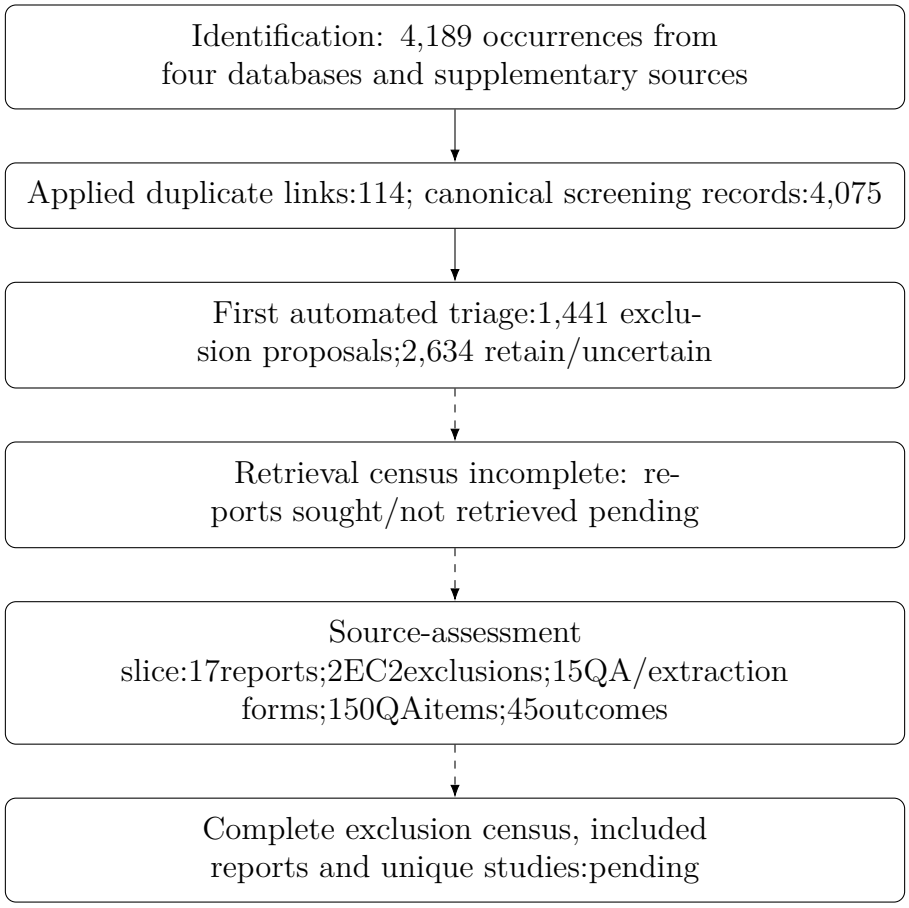
IC1-IC6 are conjunctive. At assessed full text, one decisive exclusion reason is recorded in protocol order. A report that has not been retrieved is not an EC6 assessed exclusion.

IC	Required condition
IC1	English peer-reviewed full primary conference/journal report; final source identity checked.
IC2	Evaluated sharing among concurrent tenants/models/processes on one physical GPU, including temporal/hybrid sharing.
IC3	Inference/model serving; mixed workloads need separable inference outcomes.
IC4	Relevant inspectable outcome with interpretable experimental configuration.
IC5	Extractable within-one-GPU evidence and sufficient source detail; multi-GPU scale alone does not exclude.
IC6	Main publication window or a justified older seminal exception.

EC	Decisive exclusion condition
EC1	Non-primary/non-peer-reviewed version, survey, tutorial, editorial, preprint-only or abstract/poster item.
EC2	Training-only or no evaluable concurrent inference-sharing outcome.
EC3	Hardware/microarchitecture-only work without evaluated systems-level sharing.
EC4	Only aggregate cluster/multi-GPU/remote-execution results without extractable local co-tenant evidence.
EC5	Non-GPU accelerator without a relevant evaluated GPU mechanism.
EC6	Assessed full text has insufficient methods/configuration/outcomes for any RQ.
EC7	Outside period without accepted exception or non-English full text.

11 PRISMA 2020 interim accounting

The diagram is interim. Preliminary triage proposals are not final screened exclusions. Reports sought/not retrieved and final eligibility have no complete reconciled census yet. The 17 source-assessed reports comprise two EC2 exclusions and 15 QA/extraction forms. This is a source-assessment slice, not the arithmetic remainder of 2,634 candidates.



Report / decision	Source-based reason	Exact anchor
GB-P016 TGS: Transparent GPU Sharing in Container Clouds for Deep Learning Workloads; EC2	Training-focused GPU container sharing; evaluation concerns production/ opportunistic training jobs. Discussion explicitly places inference optimization outside this paper; no separable inference-sharing result identified.	printed p69 abstract; evaluation pp75-80; p81 discussion; PDF pp2,8-14
GB-P017 HiveD: Sharing a GPU Cluster for Deep Learning with Guarantees; EC2	GPU-cluster resource reservation and affinity for deep-learning training; evaluated queueing/completion-time behavior is not concurrent local inference serving. EC2 decisive; aggregate cluster scope is a secondary limitation.	printed p515 abstract; section6 evaluation pp521-525; PDF pp2,8-12

12 Search, screening, duplicate and abstract logs

The compact workbook carries 4,075 canonical screening rows and the 4,189-occurrence duplicate ledger. Screening rows contain database, year, DOI, first recommendation, reason, report/study links, full-text source/anchor and final status. Record IDs remain stable. The three merged canonical rows and original opinions are retained rather than deleted.

Whitespace/entity normalization was audited for all 4,189 abstract fields without publishing the full text or changing original exports.3,658 occurrences contain some abstract text and531 are blank. Normalization does not verify that a field is a complete abstract; snippets and truncation remain possible. The audit records character counts and source hashes rather than treating empty or incomplete text as a scientific finding.

13 Quality assessment

Group B uses ten 0-2 criteria. A reasoned N/A removes its maximum from the denominator. Below 60 percent is a sensitivity view, not an exclusion gate. All 15 assessed forms have ten item rows; none is converted to human-validated appraisal. The separate QA document supplies every rationale, anchor, assessor and remaining check.

Report	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total
GB-P001	2	2	1	1	1	2	NA	2	1	1	13/18
GB-P002	2	2	2	1	1	2	2	2	1	2	17/20
GB-P003	2	2	2	2	2	2	2	2	1	1	18/20
GB-P004	2	2	2	1	2	2	2	2	1	1	17/20
GB-P005	2	2	1	2	2	2	1	2	1	2	17/20
GB-P006	2	2	1	2	1	2	2	2	NA	2	16/18
GB-P007	2	2	2	2	1	2	2	2	2	2	19/20
GB-P008	2	2	1	2	2	2	1	2	2	1	17/20
GB-P009	2	2	1	1	1	2	2	2	1	2	16/20
GB-P010	2	2	1	2	1	2	2	2	1	2	17/20
GB-P011	2	2	1	1	1	2	1	2	1	1	14/20
GB-P012	2	2	1	1	1	2	2	2	1	1	15/20
GB-P013	2	2	1	1	1	2	1	2	1	2	15/20
GB-P014	2	2	1	2	1	2	2	2	1	1	16/20
GB-P015	2	2	1	1	2	2	2	2	NA	2	16/18

Provisional N/A judgments are VastPipe Q7, Paella Q9 and XSched Q9. Claim-specific applicability and the planned calibration remain open. Twenty legacy items have section/ figure anchors without original printed-page mapping; additional section-only anchors remain visible in QA follow-up. Current first scores produce no below 60 percent forms, but incomplete eligibility/version checks prevent a strong robustness claim.

14 Topic-specific extraction and study identities

The compact summary below links each report to its mechanism, hardware and assessment status. The separate data-extraction document and workbook retain full study forms and all

45 outcome rows. Every reported metric has a value/unit, workload, comparator and anchor. Two additional calculations retain explicit formulas and inputs and remain separate from reported measurements. Unknown fields are not extracted; a not-reported label is scoped to the source section actually checked.

GB-P001 VastPipe

[7]

Dimension	Source-linked extraction
Identity	Li-Xian Ma and Le-Ping Wang and En Shao and Rong-Yu Cao and Guang-Ming Tan; 2025; Journal of Computer Science and Technology
Mechanism / GPU	tuned/default MPS comparison; prior source check only; NVIDIA A100 (existing source-check register)
Workload / conditions	VGG16+GoogLeNet (existing extraction rows); not extracted; not extracted
Comparator	No MPS / default MPS;exact quotas not extracted
Eligibility / source	provisionally_eligible; pp457-459, sections6.7-6.9, Figs14-16

GB-P002 iGniter

[8]

Dimension	Source-linked extraction
Identity	Fei Xu and Jianian Xu and Jiabin Chen and Li Chen and Ruitao Shang and Zhi Zhou and Fangming Liu; 2023; IEEE Transactions on Parallel and Distributed Systems
Mechanism / GPU	MPS quota allocation and shadow process recovery;prior source check; NVIDIA V100 (existing source-check register)
Workload / conditions	AlexNet;ResNet50;VGG19;SSD; not extracted; batch3 for prediction check;W1 SLO10ms for recovery example
Comparator	Measured latency for prediction error;10ms SLO for recovery
Eligibility / source	provisionally_eligible; pp820-824, sections4.2,5.1-5.4, Figs13-17

GB-P003 gpulets

[9]

Dimension	Source-linked extraction
Identity	Seungbeom Choi et al.; 2022; USENIX ATC22
Mechanism / GPU	MPS spatial partitions plus temporal batching; interference-aware gpulet scheduler; RTX2080Ti11GB; two GPUs/server, two servers; local pair measurements separated

Dimension	Source-linked extraction
Workload / conditions	GoogLeNet,LeNet,ResNet50,SSD-MobileNet,VGG16,MnasNet,MobileNetv2,DenseNet,BaseBERT (Table3); paired local inferences; system scenarios contain several models; Poisson inter-arrivals; model SLO5-202ms;99% requests within SLO; median of3 runs (pp207-208)
Comparator	Solo latency for local Fig7; SBP and greedy best-fit for system scheduling
Eligibility / source	provisionally_eligible; pp201-203 local mechanism/Fig7; p207 Tables2-3;pp208-209 system methodology

GB-P004 PipeSwitch

[10]

Dimension	Source-linked extraction
Identity	Zhihao Bai;Zhen Zhang;Yibo Zhu;Xin Jin; 2020; USENIX OSDI20
Mechanism / GPU	Temporal task switching; pipelined model transfer; active/standby workers and shared memory daemon; V10016GB PCIe3x16;T416GB PCIe3x8
Workload / conditions	ResNet152;Inception_v3;Bert_base; one training task preempted for one inference task; separable inference latency; training32;inference8;100-run averages;Fig6 reports minimum/ maximum (p508)
Comparator	Ready model;stop-and-start;MPS using unified memory
Eligibility / source	provisionally_eligible; p508 section6/6.1; p509 Tables3-4

GB-P005 REEF conference

[11]

Dimension	Source-linked extraction
Identity	Mingcong Han;Hanze Zhang;Rong Chen;Haibo Chen; 2022; USENIX OSDI22
Mechanism / GPU	Reset-based preemption;idempotent inference kernels;dynamic kernel padding;runtime/driver changes; AMD Radeon Instinct MI50:60CUs/ 16GB;REEF-N extension NVIDIA V100
Workload / conditions	ResNet152;DenseNet201;VGG19;Inceptionv3;DistilBERT; DISB A/B: 1RT+1BE;C:1RT+5BE;D/E:5RT+5BE; Table2 RT100/220/100/20/ 20req/s;D uniform,E Poisson;BE closed-loop;batch not extracted
Comparator	RT-Only;SEQ;GPUStreams;separate wait-based preemption microbenchmark
Eligibility / source	eligible_companion_candidate; p548 section7.1 Table2;p549 section7.2 Fig11;section8 idempotence limitation

GB-P006 Paella: Low-latency Model Serving with Software-defined GPU Scheduling

[12]

Dimension	Source-linked extraction
Identity	Kelvin K.W. Ng; Henri Maxime Demoulin; Vincent Liu; 2023; SOSP 2023
Mechanism / GPU	Instrumented kernels; software scheduling; Tesla T4
Workload / conditions	Mixed inference; lognormal arrivals sigma1.5/2; Mixed inference; lognormal arrivals sigma1.5/2; not extracted
Comparator	CUDA-MS;Triton23.03;Paella ablations
Eligibility / source	provisionally_eligible;study_identity_provisional; printed pp604-606; PDF pp10-12; sections7.1-7.3

GB-P007 Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications

[13]

Dimension	Source-linked extraction
Identity	Foteini Strati; Xianzhe Ma; Ana Klimovic; 2024; EuroSys 2024
Mechanism / GPU	Interception; interference-aware kernel scheduling; V10016GB;A10040GB extension
Workload / conditions	High-priority plus best-effort inference;Poisson; High-priority plus best-effort inference;Poisson; not extracted
Comparator	Dedicated high-priority GPU;MPS;Streams;REEF-N policy
Eligibility / source	provisionally_eligible;study_identity_provisional; printed pp1082-1087; PDF pp8-13; sections6-7

GB-P008 GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform

[14]

Dimension	Source-linked extraction
Identity	Aditya Dhakal; Sameer G Kulkarni; K.K. Ramakrishnan; 2020; SoCC 2020
Mechanism / GPU	MPS allocation; adaptive batching; One V10016GB
Workload / conditions	Concurrent inference functions;25ms target; Concurrent inference functions;25ms target; not extracted
Comparator	Default MPS;MPS+SLAB;TensorRT19.02
Eligibility / source	provisionally_eligible;study_identity_provisional; printed pp499-501; PDF pp8-10; sections4.1-4.3

GB-P009 Serving DNNs like Clockwork: Performance Predictability from the Bottom Up

[15]

Dimension	Source-linked extraction
Identity	Arpan Gujarati; Reza Karimi; Safya Alzayat; Wei Hao; Antoine Kaufmann; Ymir Vigfusson; Jonathan Mace; 2020; USENIX OSDI 2020
Mechanism / GPU	Predictable temporal scheduling; One V10032GB in section6.1
Workload / conditions	15 ResNet50 instances;16 closed-loop clients/model; 15 ResNet50 instances;16 closed-loop clients/model; not extracted
Comparator	Clipper;INFaaS (different EC2 hardware)
Eligibility / source	provisionally_eligible;study_identity_provisional; printed pp451-452; author PDF pp9-10; section6.1/Fig5

GB-P010 Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS)

[16]

Dimension	Source-linked extraction
Identity	Jiali Wang; Yankui Wang; Mingcong Han; Rong Chen; 2025; USENIX ATC2025, pp1657-1675
Mechanism / GPU	hybrid spatial sharing; CUDA VMM memory handover and dynamic SM masking; separate inference/training runtimes; single NVIDIA Tesla V100 16GB in section6.1; four-V100 host; separate section6.6 single A100 80GB LLM experiment
Workload / conditions	ResNet152 319MB; DenseNet169 84MB; EfficientNetV2-s86MB; EfficientViT-b2 109MB; DistilBERT254MB; DistilGPT2 317MB; Swin-T training; section6.6 Llama2-13B26GB with Qwen2-0.5B training; six inference models duplicated round-robin to56 instances; co-located Swin-T training, batch72; Light/Heavy/Burst/Skewed and compressed Azure Functions traces; 300s runs; SLO=4x standalone inference execution; Tidle5s, watermark1GB
Comparator	TaskSwitch implemented by authors, not vanilla PipeSwitch; SP-50/ SP-75 static memory; UM+MPS; Infer-Only
Eligibility / source	provisionally_eligible_AI_source_assessment; printed pp1665-1666, section6/6.1, Table3, Figure10; PDF pp10-11 (includes cover)

GB-P011 ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments

[17]

Dimension	Source-linked extraction
Identity	Munkyu Lee; Sihoon Seong; Minki Kang; Jihyuk Lee; Gap-Joo Na; In-Geol Chun; Dimitrios Nikolopoulos; Cheol-Ho Hong; 2024; SC2024 identity; extracted text is arXiv2409.14447v1
Mechanism / GPU	hybrid MIG+MPS; MPS processes within a MIG segment; profiler/ configurator plus fleet allocator; A10080GB; Amazon p4de.24xlarge hosts with8 GPUs; extracted profiler point concerns one MIG segment
Workload / conditions	11 DNN workloads overall; local extracted example InceptionV3; one to three homogeneous MPS processes within a MIG instance; other models can occupy separate segments; extracted example:1-GPC MIG instance; batch4;1/2/3 MPS processes; no tail quantile identified for local latency
Comparator	same InceptionV3/MIG/batch at1 MPS process versus2/3; fleet baselines not used for local effect
Eligibility / source	uncertain_publication_version;technical_scope_supported; author preprint PDF pp4-5 Figures3-4; p8 experimental environment

GB-P012 Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference

[18]

Dimension	Source-linked extraction
Identity	Minchen Yu; Ao Wang; Dong Chen; Haoxuan Yu; Xiaonan Luo; Zhuohao Li; Wei Wang; Ruichuan Chen; Dapeng Nie; Haoran Yang; Yu Ding; 2025; USENIX ATC2025 pp597-612
Mechanism / GPU	API redirection/remoting plus temporal request-level sharing and host/GPU model swapping; GPU runtime sharing mode; single V10032GB in local Figure8; evaluation worker has4 V10032GB GPUs
Workload / conditions	ResNet152 local sharing test;8 models in Table4 including DenseNet169/201,InceptionV3,EfficientNet,ResNet50/101/ 152,Bert-qa; multiple low-frequency ResNet152 functions share one GPU; sufficient functions to saturate device memory; 10-80 requests/ minute/function; global defaultP98; CV deadline80ms and Bert-qa200ms
Comparator	Native early binding; local remoting microbenchmark also compares GVirtuS; cluster INFless-KA results kept separate
Eligibility / source	provisionally_eligible_AI_source_assessment; pp604-606 sections7.1-7.2, Table4, Figure8; PDF pp9-11

GB-P013 StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow

[19]

Dimension	Source-linked extraction
Identity	Hao Wu; Yue Yu; Junxiao Deng; Shadi Ibrahim; Song Wu; Hao Fan; Ziyue Cheng; Hai Jin; 2024; USENIX ATC2024 pp59-73
Mechanism / GPU	API forwarding into workflow-shared CUDA runtime/streams; PTB software spatial partitioning and fine-grained memory pool; V10016GB,80CUs; host has4 GPUs; single-GPU multi-user test in section9.4
Workload / conditions	Image Processing,Traffic,Social Media workflows: SSD300,FSRCNN,ResNet152/50/18,VGG16,AlexNet,LSTM,BERT; concurrent functions within workflow share runtime; separate3/6-user Traffic test on one GPU; Azure Functions7-day invocation trace; sporadic/periodic/bursty; Traffic SLO200ms; up to3 workflows/ runtime contexts per GPU in evaluated setup
Comparator	INFless and Astraea with MPS; Stream-only; all warm-up/ model-sharing/pipelined-loading settings disclosed in section9
Eligibility / source	provisionally_eligible_AI_source_assessment; pp67-70 sections9.1-9.4; Figures12-14,19; PDF pp10-13

GB-P014 Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON)

[20]

Dimension	Source-linked extraction
Identity	Shulai Zhang; Ao Xu; Quan Chen; Han Zhao; Weihao Cui; Zhen Wang; Yan Li; Limin Xiao; Minyi Guo; 2025; USENIX ATC2025 pp1003-1019
Mechanism / GPU	kernel-command-buffer interception; feedback-controlled temporal quotas and MIG spatial orchestration; Linux loadable module; A10040GB for local quota/QoS tests; twoRTX4090 used for separate distributed-training test
Workload / conditions	local Transformer-XL inference batch8/16/32;ResNet50/ MobileNetV2/BERT-Large also benchmarked; inference with co-running applications; separate training/rendering results not substituted; QoS:Poisson arrivals at half maximum throughput;50%temporal allocation;5/10/20/40/80ms tokens; target original soloP99
Comparator	original soloP99 QoS target; fixed temporal quotas; adapted MIG baselines used separately for fleet packing
Eligibility / source	provisionally_eligible_AI_source_assessment; pp1011-1015 sections5.1,5.5-5.7,7; PDF pp10-14

GB-P015 XSched: Preemptive Scheduling for Diverse XPUs

[21]

Dimension	Source-linked extraction
Identity	Weihang Shen; Mingcong Han; Jialong Liu; Rong Chen; Haibo Chen; 2025; USENIX OSDI2025 pp671-692
Mechanism / GPU	temporal/preemptive runtime interposition via XQueue/XShim; hardware-specific preemption levels; single NVIDIA GV100 for extracted Triton/Paella case; NPU/ASIC/FPGA results outside extracted GPU outcomes
Workload / conditions	two Bert-large model clients for Triton; Paella mixed-model request workload for second case; one high-priority client at10requests/ s;low-priority client continuous on sameGV100; Triton client rates as above;Paella workload lognormal sigma2.0;K-EDF K16; exact batch sizes not reported in extracted case3
Comparator	vanilla Triton;Triton priority setting;standalone high-priority execution;separate comparison to Paella
Eligibility / source	provisionally_eligible_AI_source_assessment; p683 section7.4 case3/ Figure15; PDF p14

15 Taxonomy and integration layers

Family	Assessed evidence / boundary
Temporal/preemptive	PipeSwitch,REEF,Clockwork,Paella,Orion,XSched andTorpor contain different scheduling mechanisms; their priority, switching and batching costs are not interchangeable.
MPS concurrency	VastPipe,iGniter,gpulets andGSLICE use or compare local spatial concurrency; allocation and model mix change interference.
MIG / hybrid	KRYPTON integrates MIG with time quotas;ParvaGPU characterizes MIG+MPS but accessed version remains uncertain. No pooled hardware-isolation gain is inferred.
API interception/ remoting	Torpor redirects APIs with model swapping;StreamBox forwards APIs into workflow runtimes;KRYPTON intercepts command buffers in kernel space. Those boundaries have different trust and integration costs.
Memory-sharing hybrids	SIRIUS uses VMM handover and SM masks across inference/ training;StreamBox shares runtime/memory within a workflow.

16 Comparative evidence

Numerical comparisons are within-paper contrasts. They are not ranked across unmatched GPUs, models, batches, load generators, memory capacities or baselines. Each row retains a representative outcome; the complete 45-row register is separate.

Report	Representative result	Comparator	Anchor / interpretation limit
GB-P001	Throughput ratio: 2.7 x	No MPS	pp457-458 Fig14 and accompanying prose; Batch/ rate/quotas unresolved; not tail-latency evidence
GB-P002	Prediction error range: 1.53-5.02 percent	Measured local latency	p822 section5.2 Fig13; Prediction accuracy, not utilization or latency speedup
GB-P003	Distribution threshold: 90% of measured pairs: 17.66 percent overhead below threshold	Solo-running latency	p203 Fig7; Across-pair CDF, not request-p90/p99; no hostile-memory/fault test
GB-P004	Additional end-to-end latency: 6.01 ms	Ready-model latency subtracted; MPS overhead307.02ms	p509 Table3; setup p508 section6; Mean100 runs; overhead, not total latency or p99; MPS uses unified-memory swapping
GB-P005	Overall throughput ratio: 1.60 x	RT-Only dedicated GPU	p549 section7.2 Fig11 prose; setup p548 Table2; Includes RT and BE requests; RT-only reference dedicates GPU to RT tasks.
GB-P006	ResNet18 p99 reduction factor: up to3 x	CUDA-MS	printed p604 section7.1/Fig12; PDF pp10-11; Author-reported approximate maximum; not uniform across loads
GB-P007	Aggregate throughput ratio: up to7.3 x	Dedicated high-priority inference	printed p1086 section6.2.3/ Fig12; PDF p12; Maximum; includes best-effort work
GB-P008	GPU utilization efficacy GUE: 94.04 composite score	Default MPS GUE38.82	printed p500 Table3; PDF p9; Not percent GPU utilization
GB-P009	Clockwork goodput: approximately800 requests/second	INFaaS on nonidentical EC2 setup	printed p452 section6.1/Fig5; author PDF p10; Within-SLO successes; comparator qualitative; not total throughput
GB-P010	Average P99 inference latency improvement factor, as reported: 12.6 x	TaskSwitch (authors' adapted temporal-switch baseline)	printed p1666, section6.1/ Figure10 discussion; PDF p11; Author-reported average across tested workloads; not a uniform per-model speedup, and not a comparison to unmodified PipeSwitch.
GB-P011	Reported throughput: 446 requests/s	same configuration with1 process (354requests/s)	author preprint PDF p5, sectionIII-B prose; Figure3 p4; One local configuration, not fleet savings. Source is arXivv1; final published version not verified.

Report	Representative result	Comparator	Anchor / interpretation limit
GB-P012	Reported single-GPU throughput factor: >10 x	Native early binding at same offered function rate	p606 section7.2; Figure8 p605; PDF pp10-11; Lower bound reported by authors, not an exact10x; function count varies to saturate memory. No fleet cost claim.
GB-P013	Author-reported average throughput factor: 6.7 x	INFless with documented warm-up/ pipelined loading/ model sharing	p68 section9.1/Figure13; PDF p11; Aggregate within-paper result across tested workflows; not universal or a matched cross-paper speedup. Batch/ concurrency optimizations contribute.
GB-P014	Reported average QoS violation rate: 1.23 percent of requests	original soloP99 threshold (1%baseline violations by definition)	p1014 section5.5/Figure17; PDF p13; Average across tested configurations; no claim that worst-case violation is1.23%. Threshold-relative metric, not latency in milliseconds.
GB-P015	Reported high-priorityP99 latency ratio: 1.07 x standalone	standalone high-priorityBert-large execution	p683 section7.4 case3/Figure15a; PDF p14; Triton+XSched;not a rawmillisecond value. NativeTriton ratio1.53x;noNPUresultincluded.

17 Engineering trade-offs

Choice	Observed benefit / required sacrifice
Tuned MPS	VastPipe throughput contrasts do not establish tail-latency or fault isolation; iGniter recovery traces retain transient SLO violation.
Preemptive priority	XSched high-priorityTritonP99 is1.07xstandalone versus1.53xnativeTriton, with a continuously busy background client; background throughput remains a separate objective.
Memory handover	SIRIUS improves local inference tails versus an adapted switch baseline while opportunistic training uses residual resources; training gains are not inference throughput.
Workflow runtime sharing	StreamBox eliminates runtime redundancy but cross-user concurrency can cause queuing and SLO violations; within-workflow address-space sharing limits isolation.
Remoting/model swapping	Torpor increases low-rate function density; heavy model transfers can increase latency sharply and add CPU/bandwidth demand.
Temporal quota granularity	KRYPTON token lengths trade scheduling overhead against wait time, differently for different batches.
MIG+MPS configuration	ParvaGPU preprint example changes throughput354to446requests/s while latency11to27ms; this is version-qualified local evidence, not a final included-study claim.

18 Conflicting evidence and comparability

Different results are not automatically contradictory. GSLICE reports a negative latency contrast against TensorRT in one configuration, while other outcomes evaluate different baselines and statistics. Torpor model swapping can hurt a heavy model even when function density improves. These within-paper differences support condition-aware choices rather than an averaged universal speedup.

The ParvaGPU local example illustrates a concrete tension: at1-GPC MIG size and batch4, moving from one to three MPS processes changes throughput354to446requests/s and latency11to27ms. The separately calculated increase is25.99percent throughput and2.4545xlatency, with explicit inputs retained. Latency percentile is unspecified and final publication equivalence is unresolved. These values are not pooled with P99/P98 request tails from other systems.

Cross-study comparisons remain qualitative where hardware, tenant mix, batching, load, runtime sharing or measurement intervals differ. No matched, controlled study-to-study contradiction or universal winning mechanism has been established by this checkpoint.

19 Answers to the research questions

RQ	Evidence-based checkpoint answer
RQ1	The assessed slice covers temporal scheduling,MPS,API interception/ remoting,memory-sharing hybrids and MIG-integrating designs. Prevalence across the literature cannot be inferred from the incomplete selected set.
RQ2	The register distinguishes requests/s,throughput ratios,P99/P98,SLO-violation rate,API overhead and host/GPU memory. These are not interchangeable measures of utilization.
RQ3	Batch,process count,arrival pattern,model size,runtime redundancy,hardware capability and memory/PCIe pressure alter outcomes. Local example settings accompany every metric.
RQ4	Scheduling fairness and resource quotas have more direct evidence than general security/fault guarantees. Compiler transformation,kernel modules,model swapping and CPU-side API handling impose different costs.
RQ5	Select a candidate mechanism by service priority and constraints, then validate it under the intended model mix and load. Final recommendation confidence is limited by incomplete eligibility,source versions and single-AI assessment.

20 Conditional engineering recommendations

Scenario	Candidate / evidence	Cost and alternative
Trusted multi-model inference with configurable quotas	Test MPS allocation approaches against native sharing on the targetGPU; VastPipe/iGniter/ gpulets/GSLICE supply condition-specific leads.	Check tail latency and memory interference. Use stronger partitioning where trust boundaries require it; do not infer security from throughput.

Scenario	Candidate / evidence	Cost and alternative
Foreground request-tail priority	Test preemptive scheduling;XSchedTriton case andREEF/Orion/Paella offer different integration choices.	Account for hardware preemption support and background slowdown; use batching/dedication when isolation/integration demands dominate.
Workflow-local function density	TestStreamBox-style runtime/ memory sharing where workflow functions share a trust boundary.	Requires compilation/profiling and controlled concurrency; separate runtimes/partitions suit stronger isolation requirements.
Many low-rate model functions	TestTorpor late binding against residentNative baselines under target model sizes.	Host memory,CPU andPCIe traffic increase; keep heavy/ high-rate models resident when transfer cost dominates.
Mixed inference and opportunistic training	TestSIRIUS-type memory handover using separable inferenceSLO metrics.	Requires runtime integration and controlled memory reclamation;dedicate resources where interference risk is unacceptable.
MIG-aware deployments	TreatKRYPTON/ParvaGPU as design leads with different source/ validation status.	Verify supportedpro-files,reconfiguration limits and finalsource;hardware partitions alone do not establish a measured end-to-endsecurity guarantee.

21 Research gaps

The checked set does not establish matched comparisons across all mechanisms on one common workload and hardware stack. Energy,security/fault tests,uncertainty intervals and software-version detail are uneven. The final report-to-study map,source-version checks and fullcandidate coverage remain incomplete. These gaps limit confidence before any deployment recommendation is generalized.

22 Threats to validity and limitations

Threat	Handling / residual limitation
Search reproducibility	Executed versus planned queries remain distinct; unresolved ACM/ScienceDirect totals and field/filter deviations are disclosed.
Selection bias	All records have initial automated triage, but 2,634 retain/ uncertain records are not fully assessed. Accessible sources are a biased slice.
Reviewer judgment	AI-only continuation is disclosed. TwoAIopinions are not independent human validation; original scores/votes are preserved.

Threat	Handling / residual limitation
Source/version identity	ParvaGPUpreprint andREEFcompanion identity remain explicit. No inferred DOI substitution or automatic version merge.
Measurement interpretation	P99,P98,mean/unqualifiedlatency,model-pairpercentiles,training throughput and host-memory overhead remain separate.
Incomplete extraction	Not-extracted fields and section-scoped absence reporting remain visible; missing values are not zeros.
Final accounting	The final included-study count and final PRISMA stages remain pending; 15 forms do not meet the final-study target by themselves.

The submitted Task 1 protocol is byte-for-byte preserved. The current workbook, PDFs and supporting CSV files describe the same current register checkpoint. This is an updated evidence report with explicit unfinished review work, not a claim that every candidate has completed full-text eligibility.

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